The Leslie Matrix for population growth

An application of linear algebra

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Why care about population growth?

Population growth is a topic many people care about. Ecologists, biologists, urban planners, and many other people may want to answer questions like:

- How many people will be living in my city in 50 years?
- How fast will the bacteria on my Petri dish grow?
- Is this species in danger of going extinct?

What does linear algebra have to do with population growth?

Short answer: everything! The fundamental objects we use to model populations are vectors and matrices. In the next few slides, we will introduce a very specific kind of matrix and use it to predict an example population. We will then move on to predicting how that population will behave in the far future.

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Assumptions

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- the number of males is a constant function of the number of females (we'll only model the female part of the population)
- individuals only enter or leave the population through birth and death
- the population is separated into *age classes* (i.e. infant, juvenile, adult)

Parameters

With these assumptions, we only need three parameters to model our population! First:

Definition

- Let a_i be the average number of offspring born to a female in the ith age class, and

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- Let a_i be the average number of offspring born to a female in the ith age class, and
- Let b_i be the average proportion of females in the ith age class who survive to the $i+1$th age class.

Parameters

Additionally, we need to know where our population starts. A good way to store this information is a vector:

Definition

Let the entry $x_i^{(k)}$ of $\vec{x}^{(k)} = \begin{bmatrix} x_1^{(k)} \\ x_2^{(k)} \\ \vdots \\ x_n^{(k)} \end{bmatrix}$ represent the number of individuals in the ith age class during the kth generation of a population. We call $\vec{x}^{(k)}$ the *age distribution vector* of a population at generation k.

And this is all we need to get started!

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Some definitions

Definition

We define the *Leslie Matrix* of a population to be the matrix L such that $L\vec{x}^{(k)} = \vec{x}^{(k+1)}$ for the age distribution vector at any generation k.

Constructing a Leslie Matrix

Recall that for an $m \times n$ matrix $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \dots & \vec{a}_n \end{bmatrix}$ and an n-entry vector $\vec{v}$ with entries $v_1, v_2, \dots, v_n$,

$$A\vec{v} = v_1\vec{a}_1 + v_2\vec{a}_2 \cdots + v_n\vec{a}_n.$$

So entries in the jth column of A get multiplied by the jth entry of input vector $\vec{v}$, and entries in the ith row of A sum to contribute to the ith entry of output vector $A\vec{v}$.

Constructing a Leslie Matrix

Consider our parameters:

- We want a_i and b_i to be multiplied by the ith entry of the input age distribution vector, so they must go in the ith column of L

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Constructing a Leslie Matrix

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- We want a_i and b_i to be multiplied by the ith entry of the input age distribution vector, so they must go in the ith column of L
- The products involving a_i represent the numbers of new offspring, so the sum of those products should appear in the first entry of our output vector. This means they go in row 1 of L
- The product involving b_i represents individuals in the $i+1$th generation, b_i should go in row $i+1$ of L

Constructing a Leslie Matrix

So a_i goes in the 1st row, ith column; and b_i goes in the ith row, $i+1$th column. Hence, we have:

Important

$$L = \begin{bmatrix} a_1 & a_2 & a_3 & \dots & a_{n-1} & a_n \\ b_1 & 0 & 0 & \dots & 0 & 0 \\ 0 & b_2 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \dots & b_{n-1} & 0 \end{bmatrix}.$$

This matrix is the basic tool we use to model population growth!

Example

Let's see how this plays out with a simple example. Consider the population described by the following table:

Age class	a_i	b_i	x_i
0-5 yrs	1	$\frac{1}{4}$	40
6-10 yrs	3	0	20

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We have $L = \begin{bmatrix} 1 & 3 \\ \frac{1}{4} & 0 \end{bmatrix}$ and $\vec{x}^{(0)} = \begin{bmatrix} 40 \\ 20 \end{bmatrix}$ as our Leslie matrix and initial age distribution vector.

Example

We can calculate subsequent age distributions by multiplying:

$$\vec{x}^{(1)} = L\vec{x}^{(0)} = \begin{bmatrix} 1 & 3 \\ \frac{1}{4} & 0 \end{bmatrix} \begin{bmatrix} 40 \\ 20 \end{bmatrix} = \begin{bmatrix} 40+60 \\ 10+0 \end{bmatrix} = \begin{bmatrix} 100 \\ 10 \end{bmatrix},$$

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$$\vec{x}^{(2)} = L\vec{x}^{(1)} = \begin{bmatrix} 1 & 3 \\ \frac{1}{4} & 0 \end{bmatrix} \begin{bmatrix} 100 \\ 10 \end{bmatrix} = \begin{bmatrix} 100+30 \\ 25+0 \end{bmatrix} = \begin{bmatrix} 130 \\ 25 \end{bmatrix},$$

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$$\vec{x}^{(3)} = L\vec{x}^{(2)} = \begin{bmatrix} 1 & 3 \\ \frac{1}{4} & 0 \end{bmatrix} \begin{bmatrix} 130 \\ 25 \end{bmatrix} = \begin{bmatrix} 130+75 \\ 32.5+0 \end{bmatrix} \approx \begin{bmatrix} 205 \\ 33 \end{bmatrix},$$

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$$\vec{x}^{(4)} = L\vec{x}^{(3)} = \begin{bmatrix} 1 & 3 \\ \frac{1}{4} & 0 \end{bmatrix} \begin{bmatrix} 205 \\ 33 \end{bmatrix} = \begin{bmatrix} 205+99 \\ 51.25+0 \end{bmatrix} \approx \begin{bmatrix} 304 \\ 51 \end{bmatrix}$$

Our population seems to be growing!

Example

We can now use Leslie matrices to "move forward" in time, predicting populations generation by generation. Our next question: is there a more sophisticated way to describe the behavior of populations in the far future?

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For example, suppose we calculate several decreasing population distribution vectors for some population. How can we tell if that population will go extinct or if it will reach an equilibrium?

Example

We can now use Leslie matrices to "move forward" in time, predicting populations generation by generation. Our next question: is there a more sophisticated way to describe the behavior of populations in the far future?

For example, suppose we calculate several decreasing population distribution vectors for some population. How can we tell if that population will go extinct or if it will reach an equilibrium?

There are some technical details beyond the scope of this presentation, but the following is a sketch of how such questions can be answered.

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Predicting behavior in the far future

We want to know how we can find a nice expression for repeated multiplication by a (Leslie) matrix. Diagonalization is an intuitive first step!

Diagonalization Review

To diagonalize a Leslie matrix, we let $P = \begin{bmatrix} \vec{v_1} & \vec{v_2} & \dots & \vec{v_n} \end{bmatrix}$ be the matrix whose columns are the eigenvectors of a Leslie matrix L, with corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Then

$$\begin{aligned} LP =& L\vec{v_1} + L\vec{v_2} + \cdots + L\vec{v_n} \\ =& \lambda_1 \vec{v_1} + \lambda_2 \vec{v_2} + \cdots + \lambda_n \vec{v_n} \\ =& P \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix}. \end{aligned}$$

Diagonalization Review

Hence,

$$LPP^{-1} = L = P \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix} P^{-1}$$

Diagonalization Reivew

To calculate powers of L, we multiply the diagonalization and it cancels nicely:

$$\left(P \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix} P^{-1} \right) \left(P \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix} P^{-1} \right)$$

$$= P \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix} (P^{-1}P) \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix} P^{-1}$$

$$= P \begin{bmatrix} \lambda_1^2 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2^2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n^2 \end{bmatrix} P^{-1}$$

Diagonalization Review

And of course, multiplying by L k times gives

$$L^k = P \begin{bmatrix} \lambda_1^k & 0 & 0 & \dots & 0 \\ 0 & \lambda_2^k & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n^k \end{bmatrix} P^{-1}.$$

We can try this out on our example.

Example revisited

Recall that we have $L = \begin{bmatrix} 1 & 3 \\ \frac{1}{4} & 0 \end{bmatrix}$ and $\vec{x}^{(0)} = \begin{bmatrix} 40 \\ 20 \end{bmatrix}$ as our Leslie matrix and initial age distribution vector.

Our game plan now:

- Find the eigenvalues and eigenvectors of L (which we need for diagonalization)
- Diagonalize L
- Use this diagonalization to describe the kth generation of the population as $k \to \infty$

Example revisited - eigenvalues

Eigenvalues λ of L satisfy:

$$\det(L - \lambda I_2) = \det\left(\begin{bmatrix} 1-\lambda & 3 \\ \frac{1}{4} & -\lambda \end{bmatrix}\right) = 0$$

Example revisited - eigenvalues

Eigenvalues λ of L satisfy:

$$\begin{aligned}\det(L - \lambda I_2) =& \det\left(\begin{bmatrix} 1-\lambda & 3 \\ \frac{1}{4} & -\lambda \end{bmatrix}\right) = 0 \\ \implies & (1-\lambda)(\lambda) - (3)(\tfrac{1}{4}) \\ & = \lambda^2 - \lambda - \tfrac{3}{4} \\ & = (\lambda - \tfrac{3}{2})(\lambda + \tfrac{1}{2}) = 0.\end{aligned}$$

Example revisited - eigenvalues

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So $\frac{3}{2}$ and $-\frac{1}{2}$ are our eigenvalues.

Example revisited - eigenvectors of L

We solve $(L - \frac{3}{2}I_2)\vec{v} = \vec{0}$ for $\vec{v}$ to find the eigenvector of $\frac{3}{2}$.

$$L - \tfrac{3}{2}I_2 = \begin{bmatrix} 1 - \frac{3}{2} & 3 \\ \frac{1}{4} & 0 - \frac{3}{2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & 3 \\ \frac{1}{4} & -\frac{3}{2} \end{bmatrix}.$$

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To row reduce, subtract $-\frac{1}{2}$(Row 1) from Row 2 and scale:

$$\begin{bmatrix} -\frac{1}{2} & 3 \\ \frac{1}{4} & -\frac{3}{2} \end{bmatrix} \to \begin{bmatrix} -\frac{1}{2} & 3 \\ 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & -6 \\ 0 & 0 \end{bmatrix}.$$

Example revisited - eigenvectors of L

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This gives $\text{Span}\left\{ \begin{bmatrix} 6 \\ 1 \end{bmatrix} \right\}$ as the set of eigenvectors corresponding to $\frac{3}{2}$. We just need one to diagonalize L; let's choose $\begin{bmatrix} 6 \\ 1 \end{bmatrix}$.

Example revisited - eigenvectors of L

Similarly, we solve $(L + \frac{1}{2}I_2)\vec{v} = \vec{0}$ to find the eigenvector of $-\frac{1}{2}$:

$$L + \tfrac{1}{2}I_2 = \begin{bmatrix} 1 + \frac{1}{2} & 3 \\ \frac{1}{4} & 0 + \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & 3 \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix}.$$

Example revisited - eigenvectors of L

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We subtract $\frac{1}{6}$(Row 1) from Row 2 and scale:

$$\begin{bmatrix} \frac{3}{2} & 3 \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix} \to \begin{bmatrix} \frac{3}{2} & 3 \\ 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

Example revisited - eigenvectors of L

Similarly, we solve $(L + \frac{1}{2}I_2)\vec{v} = \vec{0}$ to find the eigenvector of $-\frac{1}{2}$:

$$L + \tfrac{1}{2}I_2 = \begin{bmatrix} 1 + \frac{1}{2} & 3 \\ \frac{1}{4} & 0 + \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & 3 \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix}.$$

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$$\begin{bmatrix} \frac{3}{2} & 3 \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix} \to \begin{bmatrix} \frac{3}{2} & 3 \\ 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

This gives $\text{Span}\left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}$ as the set of eigenvectors corresponding to $\frac{3}{2}$. We choose $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$.

Example revisited - diagonalization of L

We wish to write $L = P \begin{bmatrix} \frac{3}{2} & 0 \\ 0 & -\frac{1}{2} \end{bmatrix} P^{-1}$, where $P = \begin{bmatrix} 6 & -2 \\ 1 & 1 \end{bmatrix}$, the matrix whose columns are the eigenvectors of L.

We find P^{-1} with row reduction:

$$\left[\begin{array}{cc|cc} 6 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 0 & -8 & 1 & -6 \\ 1 & 1 & 0 & 1 \end{array}\right]$$

$$\to \left[\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & -8 & 1 & -6 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & 1 & -\frac{1}{8} & \frac{3}{4} \end{array}\right]$$

$$\to \left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{8} & \frac{1}{4} \\ 0 & 1 & -\frac{1}{8} & \frac{3}{4} \end{array}\right] \implies P^{-1} = \begin{bmatrix} \frac{1}{8} & \frac{1}{4} \\ -\frac{1}{8} & \frac{3}{4} \end{bmatrix}.$$

Example revisited - diagonalization of L

Hence, we have

$$L = \begin{bmatrix} 6 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{3}{2} & 0 \\ 0 & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{8} & \frac{1}{4} \\ -\frac{1}{8} & \frac{3}{4} \end{bmatrix}$$

and therefore

$$L^k = \begin{bmatrix} 6 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \left(\frac{3}{2}\right)^k & 0 \\ 0 & \left(-\frac{1}{2}\right)^k \end{bmatrix} \begin{bmatrix} \frac{1}{8} & \frac{1}{4} \\ -\frac{1}{8} & \frac{3}{4} \end{bmatrix}.$$

Example revisited - the kth generation

In this example, as k grows, $\left(-\frac{1}{2}\right)^k$ shrinks towards zero, so

$$\lim_{k\to\infty} L^k = \begin{bmatrix} 6 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \left(\frac{3}{2}\right)^k & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{8} & \frac{1}{4} \\ -\frac{1}{8} & \frac{3}{4} \end{bmatrix}.$$

Example revisited - the kth generation

So, given our initial age distribution vector $\vec{x}^{(0)} = \begin{bmatrix} 40 \\ 20 \end{bmatrix}$, we approximate $\vec{x}^{(k)}$ for large k:

$$\lim_{k\to\infty} \vec{x}^{(k)} = \lim_{k\to\infty} L^k \begin{bmatrix} 40 \\ 20 \end{bmatrix} = \begin{bmatrix} 6 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \left(\frac{3}{2}\right)^k & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{8} & \frac{1}{4} \\ -\frac{1}{8} & \frac{3}{4} \end{bmatrix} \begin{bmatrix} 40 \\ 20 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \left(\frac{3}{2}\right)^k & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 5+5 \\ -5+15 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \left(\frac{3}{2}\right)^k & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \end{bmatrix},$$

Example revisited - the kth generation

$$
\begin{aligned}
&\begin{bmatrix} 6 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \left(\frac{3}{2}\right)^k & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \end{bmatrix} \\
=&\begin{bmatrix} 6 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 10\left(\frac{3}{2}\right)^k + 0 \\ 0 + 0 \end{bmatrix} \\
=&\begin{bmatrix} 6 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 10\left(\frac{3}{2}\right)^k \\ 0 \end{bmatrix} \\
=&10\left(\frac{3}{2}\right)^k \begin{bmatrix} 6 \\ 1 \end{bmatrix} \\
=&\left(\frac{3}{2}\right)^k \begin{bmatrix} 60 \\ 10 \end{bmatrix}.
\end{aligned}
$$

Predicting behavior in the far future

This means that as time goes on, our population approaches a state in which it grows 50% each generation! We can conclude that the population will keep growing forever.

(Note how this growth is acting on an eigenvector of our Leslie matrix, $\begin{bmatrix} 60 \\ 10 \end{bmatrix}$, which means that the future of our population is fundamentally related to its Leslie matrix, regardless of our initial age distribution vector. Very cool!)

Generalization

So, diagonalization answered our questions about our example population. But now we ask: what if there were more than one eigenvalue λ of L with $|\lambda| > 1$? Taking the limit as $k \to \infty$ of our diagonalization would not be nearly as nice, because such eigenvalues would not "automatically" go to zero in our diagonal matrix.

Generalization

Well, with that question in mind, we can now fully appreciate the usefulness of the following fact:

Fact

It turns out that any Leslie matrix L has a unique positive real eigenvalue that is dominant. (Arbitrarily call it λ_1.) Furthermore, if two successive entries a_i and a_{i+1} in the first row of L are nonzero, λ_1 is strictly dominant. That is, for such matrices, $\lambda_1 > |\lambda_i|$ for $i \in \{2, ..., n\}$.

Generalization

The proof of the above fact is beyond the scope of this presentation. But it means that, given any Leslie Matrix with two successive nonzero entries a_i and a_{i+1}, we can write

$$L^k = P \begin{bmatrix} \lambda_1^k & 0 & 0 & \dots & 0 \\ 0 & \lambda_2^k & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n^k \end{bmatrix} P^{-1},$$

and then divide by our dominant eigenvector (arbitrarily call it λ_1) to get

$$\frac{L^k}{\lambda_1^k} = P \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & \left(\frac{\lambda_2}{\lambda_1}\right)^k & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & \left(\frac{\lambda_n}{\lambda_1}\right)^k \end{bmatrix} P^{-1}.$$

Generalization

Now our fact tells us that $|\frac{\lambda_2}{\lambda_1}|, |\frac{\lambda_3}{\lambda_1}|, \ldots, |\frac{\lambda_n}{\lambda_1}| < 1$, which means that

$$\lim_{k\to\infty} \frac{L^k}{\lambda_1^k} = \lim_{k\to\infty} P \begin{bmatrix} 1 & 0 & 0 & \ldots & 0 \\ 0 & \left(\frac{\lambda_2}{\lambda_1}\right)^k & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \ldots & \left(\frac{\lambda_n}{\lambda_1}\right)^k \end{bmatrix} P^{-1}$$

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$$= P \begin{bmatrix} 1 & 0 & 0 & \ldots & 0 \\ 0 & 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \ldots & 0 \end{bmatrix} P^{-1}$$

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Now our fact tells us that $|\frac{\lambda_2}{\lambda_1}|, |\frac{\lambda_3}{\lambda_1}|, \ldots, |\frac{\lambda_n}{\lambda_1}| < 1$, which means that

$$\lim_{k\to\infty} \frac{L^k}{\lambda_1^k} = \lim_{k\to\infty} P \begin{bmatrix} 1 & 0 & 0 & \ldots & 0 \\ 0 & \left(\frac{\lambda_2}{\lambda_1}\right)^k & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \ldots & \left(\frac{\lambda_n}{\lambda_1}\right)^k \end{bmatrix} P^{-1}$$

$$= P \begin{bmatrix} 1 & 0 & 0 & \ldots & 0 \\ 0 & 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \ldots & 0 \end{bmatrix} P^{-1}$$

$$\implies \lim_{k\to\infty} L^k = \lambda_1^k P \begin{bmatrix} 1 & 0 & 0 & \ldots & 0 \\ 0 & 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \ldots & 0 \end{bmatrix} P^{-1}.$$

Generalization

So, to estimate an age distribution vector far in the future, we have

$$\vec{x}^{(k)} \approx \lim_{k\to\infty} L^k \vec{x}^{(0)} = \lambda_1^k P \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix} P^{-1} \vec{x}^{(0)}.$$

And as in our example, this will simplify to $c\lambda_1^k \vec{v}_1$, where $\vec{v}_1$ is the eigenvector corresponding to the dominant eigenvector λ_1, and therefore the first column of P. *This means that the nice behavior in our example generalizes to any Leslie matrix with two successive nonzero values in its first row.*

Thank you!

We have now seen the basic ideas behind modeling population growth. Thank you for your attention! Questions?

References:

Rorres, C. & Anton, H. (1979). Applications of Linear Algebra. John Wiley & Sons, Inc.

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